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    for next_state, action in get_successors(current_state, rows, cols):
        if next_state not in visited:
            visited.add(next_state)
            queue.append((next_state, path + [(action, next_state)]))

    return None

if __name__ == '__main__':
    room_grid = [[1, 0], [1, 1]]
    vacuum_start_row = 0
    vacuum_start_col = 0

    print("Initial Room State:")
    for row in room_grid:
        print(row)
    print("Vacuum starts at:", (vacuum_start_row, vacuum_start_col))
    print("-" * 30)

    solution_path = solve_vacuum_cleaner(vacuum_start_row, vacuum_start_col, room_grid)

    if solution_path:
        for step_idx, (action_taken, resulting_state) in enumerate(solution_path):
            r, c, current_grid = resulting_state
            print("Step", step_idx + 1, "- Action:", action_taken)
            for row in current_grid:
                print(list(row))
            print("Vacuum now at:", (r, c))
            print("-" * 30)
    else:
        print("No solution found to clean the room.")

```